A prospective study of dietary calcium, dairy products and prostate cancer risk (Finland).

High dietary intakes of calcium and dairy products have been hypothesized to enhance prostate cancer risk, but available prospective data regarding these associations are inconsistent. We examined dietary intakes of calcium and dairy products in relation to risk of prostate cancer in the Alpha-Tocopherol, Beta-Carotene (ATBC) Cancer Prevention Study, a cohort of 29,133 male smokers aged 50-69 years at study entry. Dietary intake was assessed at baseline using a validated 276-item food use questionnaire. Cox proportional hazards regression was used to adjust for known or suspected risk factors for prostate cancer. During 17 years of follow-up, we ascertained 1,267 incident cases of prostate cancer. High versus low intake of dietary calcium was associated with a marked increase in prostate cancer risk. The multivariate relative risk (RR) of prostate cancer for > or =2,000 mg/day compared to <1,000 mg/day of calcium intake was 1.63 (95% confidence interval (CI), 1.27-2.10; p trend < 0.0001). Total dairy intake was also positively associated with risk of prostate cancer. The multivariate RR of prostate cancer comparing extreme quintiles of intake was 1.26 (95% CI, 1.04-1.51; p trend = 0.03). However, no association with total dairy intake remained after we adjusted for calcium (p trend = 0.17). Findings were similar by stage and grade of prostate cancer. The results from this large prospective study suggest that intake of calcium or some related component contained in dairy foods is associated with increased prostate cancer risk.